

AEROTHERMODYNAMIC ASSESSMENT OF ATMOSPHERIC EMISSIONS FROM RE-ENTRY DEMISE

James Beck¹, James Merrifield², Martin Spel³, Tobias Lips⁴, Slimane Bekki⁵

¹Belstead Research Limited, Ashford, UK,

²Fluid Gravity Engineering Limited, Emsworth, UK,

³R.Tech Sarl, Verniolle, France,

⁴HTG, Bovenden, Germany,

⁵Varuna, London, UK.

ABSTRACT

An assessment of the atmospheric emissions from spacecraft and upper stage re-entries has been performed. An emissions map has been constructed accounting for the expected mass losses from simulation codes, with additional consideration for the approximations of materials which are commonly used. Further, the sizes of the produced particulate emissions have been estimated and found to be sufficiently high that the residence time is short enough that climate impact is not expected. Production of chlorine radicals from cyanate ester resins and nitrogen oxide from shock layer chemistry have also been identified as emissions of interest. Full 2D altitude-along track maps of emissions for representative re-entries have been produced for use in atmosphere models.

Index Terms— Re-entry, atmospheric emissions

1. INTRODUCTION

With the increase in space debris, and the associated mitigation requirements, the numbers of re-entering spacecraft can be expected to increase. Therefore, it is of interest to determine whether the materials emitted in the atmospheric demise of re-entering spacecraft can provide a hazard in terms of atmospheric impact, notably stratospheric ozone depletion and climate warming. In order to support this assessment, it is necessary to produce an aerothermodynamic assessment of spacecraft demise, producing a set of gaseous and particulate emissions which could be expected from typical re-entry events of spacecraft and upper stages.

As a satellite re-enters the atmosphere, the main demise of the vehicle mass occurs once the satellite has fragmented into its separate components. Due to the high entry speed of the spacecraft, 7.8km being typical for a re-entry from Low Earth Orbit, a strong shock wave forms in front of the vehicle. In this shock wave, the air is heated to high temperatures, thereby causing the satellite components to be heated and undergo chemical changes such as thermal decomposition, oxidation and melting. As there are large uncertainties in the

fragmentation process, and the physics is not well captured by re-entry prediction tools, there is a relatively wide range of altitudes of interest for the release of gases and particulates into the atmosphere.

The high temperature chemistry during re-entry can be relatively complex. Indeed, much of the re-entry thermal processes of materials at high temperature is understood from the design of heatshields for re-entry vehicles. These cover a range of materials such as metals, ceramics and phenolic-impregnated carbon composites. However, a significant number of spacecraft materials have not been characterized during re-entry and the associated high temperature chemistry is not well known.

Importantly, many of the standard destructive re-entry codes make an assumption that all materials behave as equivalent metals, with a melting point and latent heat. For many spacecraft materials, such as CFRP, electronic circuit boards and optical glasses this model is not appropriate, and does not provide an accurate profile of the mass loss. Indeed, for any material in which the chemical reactions are important in the production of the species, some effort is required to transfer mass loss rates from an equivalent metal approximated demise to those more representative of a chemically reacting demise. This is a key aspect in the achievement of a physically sensible distribution of gaseous and particulate ablation products.

As the re-entry proceeds, the pressures in the shock layer are, in general, sufficiently low that not all chemical processes are able to reach equilibrium prior to the species being swept into the wake where they are able to cool, and the chemistry freezes. It is known that for the thermal decomposition of heatshields that, although the high temperature equilibrium suggests only small molecules will remain, laboratory tests have shown that traces of large molecules are also found.

Therefore, there is a non-equilibrium chemistry aspect, where these large molecules can interact with radiation from the sun, to create radicals which have the potential to react with ozone. It should be noted that the mass fractions of these large molecules are sufficiently small that this is not expected to have a significant impact. Indeed, the CFD simulations

performed suggest that the chemistry in the wake is sufficient that these larger molecules would not remain present in any meaningful quantities.

As the spacecraft fragments are slowed by the atmosphere, the strength of the shock reduces, resulting in the shock-layer temperatures falling, eventually to the point at which the reactions cease. The peak pressures in the shock layer are of the order of 0.1bar, but this occurs well below the peak heating point, at the stage where the chemistry is stopping. Therefore, most of the gases and particulates are produced before this occurs. The vast majority of the chemistry can be expected to be completed by an altitude of approximately 50km.

The species of interest produced by each fragment depend on the materials composing the fragment. Metallic objects are expected to produce a combination of metal and metal oxide particulates. Aluminium, especially, has a very robust oxide, which is able to protect the metal such that relatively large oxide-coated metal droplets can be formed. The expected maximum sizes can be determined from the expected shear environment using Weber number correlations as used in spray breakup modelling. Titanium, on the other hand, produces a very flaky oxide which would be expected to be removed from the surface. These particulates, especially aluminium oxide, have been shown to provide surfaces where the destruction of ozone can be enhanced.

Ceramic materials, such as silicon carbide, can be very robust to re-entry heating, but can undergo active oxidation where gaseous products are formed. Composites are more complex; these materials generally undergo endothermic thermal degradation, releasing pyrolysis gases into the shock layer. These gases tend to break down into smaller molecules, but some large molecules and hydrocarbons could survive for a limited time. There is also the possibility of oxides of nitrogen (NO_x) being produced, which could be of interest.

As well as the species produced in the shock layer, there is the possibility of some further chemistry within the hot, but cooler wake region. This wake is relatively small for fragments once they have been separated from the spacecraft. Non-equilibrium chemically reacting CFD codes can be used to augment a basic equilibrium analysis in these regions.

For complex materials, which can be toxic, the chemistry at very high temperature is not generally known. Therefore, the most reasonable approach is to start with an equilibrium analysis. For residual propellant, such as hydrazine, an analysis can be made assuming incomplete combustion and some molecular breakdown due to the high temperature.

2. EMISSIONS OF INTEREST

The number of chemical species present on spacecraft and upper stages is very high, and the uncertainties surrounding the aerochemistry during destructive re-entry are very large. Therefore, it would be possible to perform a substantial

amount of work in providing an assessment of the re-entry chemistry. In order to make this problem tractable for this activity, some selection of the materials, and chemical pathways of interest, must be established to drive the analysis.

From an environmental impact perspective, the key interest is in ozone depletion, greenhouse emissions and potential toxins. The list of species which are of particular interest is given in Table 1. It is noted that intermediate products such as organic materials, ammonia and methanoic acid are considered to be quickly oxidised in the climate modelling, such that only the final products are relevant. The separation of the carbonaceous species into soot, carbon monoxide and carbon dioxide is also of interest.

Species	Impact	Timescales
Chlorine	Stratospheric ozone	2-3 years
Nitrogen oxides	Stratospheric ozone	2-3 years
Carbon monoxide	Stratospheric ozone	~1 month
Water	Stratospheric ozone, climate	2-3 years
Hydrogen	Stratospheric ozone	2-3 years
Methane	Stratospheric ozone	2-3 years
Carbon (soot)	Stratospheric ozone, climate	<1 week to 2-3 years (depends on size)
Metal/metal oxide	Stratospheric ozone, climate	<1 week to 2-3 years (depends on size)
Carbon Dioxide	Climate	>100 years

Table 1: Relevant Species for Environmental Impact

From this data, it is simplest to consider three categories of emission; particulates, gaseous products and toxic species. It is useful to separate the emissions into these three categories, and to identify which spacecraft materials are likely to provide each product. Therefore, the analysis of the particulate and gaseous species producing materials will be separated, with specific attention paid to species which have high initial masses and/or produce small particles or particular products of interest.

3. REPRESENTATIVE VEHICLES

Two distinct types of re-entering object are considered; spacecraft and upper stages. The vast majority of spacecraft which currently re-enter do so in an uncontrolled manner. The spacecraft is on a gradually decaying circular trajectory, resulting in a very shallow re-entry. This has the effect of producing a large debris footprint, and potentially dispersing the emissions produced over a relatively large horizontal area, as well as in altitude. Spacecraft which are sufficiently massive, or are deemed to produce a sufficiently large ground casualty risk, are required to perform a controlled re-entry. This is generally significantly steeper and produces a much smaller footprint and the emissions will be less dispersed in space as a result. In general, upper stages are designed to have a direct re-entry and the resultant trajectories are essentially similar to those of controlled spacecraft re-entries. As spacecraft and upper stages have different material make-ups and the different re-entry types have different altitude profiles

of demise, each vehicle type is required to be considered on each trajectory type.

In order to provide a complete map of the potential emissions in re-entry, the following steps are required:

- Construction of a mass loss map for each vehicle on each trajectory type. This is based on simulations from two of the major destructive re-entry tools: SCARAB and SAM.
- Determination of the relevant parameters for the emissions. For particles, this is the particle size, and for gaseous products, this based on the equilibrium chemical composition.
- Capture of shock layer chemistry, in particular the production of nitrogen oxides in the shock layer. This work is performed using the MISTRAL CFD code.
- Production of a full emissions map inclusive of altitude and along track distance, accounting for variation in the re-entry vehicle, for each case.

Throughout this process, there are large uncertainties, and a gently conservative approach has been maintained throughout in order to reduce any risk of underestimating the climate impact.

The material make-up of the vehicles was provided by ESA. A number of the materials on the spacecraft are not typically simulated in destructive re-entry codes, and often proxy materials for electronics and battery materials are used. Therefore, a more careful representation of the material make-up of these components is needed. The data in Table 2 is used.

Material	Electronic Component Mass Fraction	Lithium-Ion Battery Mass Fraction	SCARAB El-Mat	SCARAB Bat-Mat
GFRP	0.32		0.68	
Ceramic (alumina)	0.35			
Aluminium	0.25	0.19	0.13	0.40
Copper	0.05	0.11	0.09	0.26
Steel / Iron		0.02	0.06	0.24
Nickel			0.03	
Zinc			0.02	
Tin	0.01		0.01	
Lithium		0.33 (LiMn2O4)		0.10
Carbon		0.15		
Ethylene carbonate		0.05		
Dimethyl carbonate		0.05		
Polypropylene		0.10		
Silicon	0.02			

Table 2: Material Composition for Electronics and Batteries

Material	Mass Fraction
Aluminium	0.346
Titanium	0.04
Steel	0.06
Copper	0.075
Glass	0.02
Cyanate Ester Resin	0.10
Carbon (incl. Fibres)	0.15
PET / PTFE / PFA / Polypropylene	0.064
GFRP	0.064
Alumina	0.07
LiMn2O4	0.011
TOTAL	1.000

Table 3: Representative Spacecraft Material Composition

The data provided by ESA was filtered to capture the particulate producing and relevant gaseous emission

producing species. Minor species, and species which do not produce emissions of interest were removed. The resultant mass composition for the representative spacecraft is given in Table 3, and the resultant mass composition for the representative upper stage in Table 4.

Material	Mass Fraction
Aluminium	0.585
Steel	0.035
Nickel/Inconel	0.1
Cyanate Ester Resin	0.068
Carbon (incl. Fibres)	0.112
PET / PU foam	0.051
GFRP	0.01
Alumina	0.008
Potassium silicate	0.024
Cork-Phenolic	0.007
TOTAL	1.000

Table 4: Representative Upper Stage Material Composition

4. MASS LOSS MAPPING

The mass loss from the re-entering spacecraft or upper stage is mapped along a range of trajectories for both uncontrolled and controlled re-entries from Low Earth Orbit. In order to achieve this, data from previous SCARAB simulations for the Sentinel fleet of spacecraft, EarthCare, CarbonSat, EPS and AVUM were used in order to obtain a map of the mass loss for each species type of interest to the emissions calculation.

The SCARAB runs suggest that about 86% of the mass in an uncontrolled re-entry becomes emission, and about 23% in a controlled re-entry, as shown in Table 5.

Material	Satellites		Upper Stages	
	Ground Fraction Of Material	Emitted Fraction Of Mass Satellite	Ground Fraction Of Material	Emitted Fraction Of Mass UpperStage
aluminium_alloy	0.0586	0.609	0.7822	0.116
copper	0.1053	0.091	0.9136	0.003
el-mat	0.1026	0.085	1	0
iron_alloy	0.6356	0.019	0.9695	0.004
cfrp	0.0473	0.038	0.3238	0.082
titanium_alloy	0.7978	0.006	0.9579	0.005
sic	0.9999	<0.001	-	-
bat-mat	0.2481	0.008	-	-
tungsten_alloy	0	0.002	-	-
zerodur	0.8685	<0.001	-	-
silicon	0	<0.001	-	-
ptfe	-	-	0.1958	0.014
norcoat	-	-	0.6885	0.004
TOTAL		0.858		0.228

Table 5: Ground Mass Fractions from SCARAB

At this point, some work is required to translate the results from the specific spacecraft considered in SCARAB to the generic representative spacecraft. Furthermore, consideration of the different approximations made in the simulations, such as the use of an aluminium material for CFRP facesheets on sandwich panels, requires accounting for. Also, a number of the organic materials, glasses and ceramics are considered too small to be a casualty risk, and the mass is represented as additional structure (aluminium) mass. Thus, the material make-up of the representative spacecraft differs from the models as shown in Tables 6 and

7. The missing mass in the SCARAB simulations is in materials which are not of interest for emissions production within the representative spacecraft and upper stage.

Material	Representative Spacecraft Mass Fraction	SCARAB Spacecraft Mass Fraction
Aluminium	0.346	0.661
Titanium	0.04	0.029
Steel	0.06	0.059
Copper	0.075	0.112
Glass	0.02	0.001
Cyanate Ester Resin	0.10	0.040
Carbon (incl. Fibres)	0.15	
PET / PTFE / PFA / Polypropylene	0.064	0
GFRP	0.064	0.064
Alumina / SiC (Ceramics)	0.07	0.024
LiMn2O4	0.011	0.001
TOTAL	1.000	0.967

Table 6: Material Comparison with Representative Spacecraft

Material	Representative Upper Stage Mass Fraction	SCARAB Upper Stage Mass Fraction
Aluminium	0.585	0.539
Steel	0.035	0.134
Nickel/Inconel	0.1	
Cyanate Ester Resin	0.068	
Carbon (incl. Fibres)	0.112	0.121
PET / PU foam	0.051	0.018
GFRP	0.01	0.023
Alumina	0.008	0
Potassium silicate	0.024	0
Cork-Phenolic	0.007	0.012
TOTAL	1.000	0.847

Table 7: Material Comparison with Representative Upper Stage

In order to reconcile these differences and to provide a mass loss map for the representative vehicles a set of simple re-entry simulations have been performed using the SAM tool. This has been done using a set of simple spherical components of varying mass and ballistic coefficient. In practice, it has been found that an exact matching of the spacecraft mass is not the most effective way of replicating the behaviour of the SCARAB mass loss, and so the correlation is performed in a material-by-material manner with a different set of objects for each material. Given that the composite and plastic materials are not modelled well in any destructive re-entry code, this analysis has been restricted to metallic materials.

	Spacecraft	Upper Stage
Initial altitude (km)	120	120
Initial latitude (deg)	45.2	45.2
Initial longitude (deg)	10.0	10.0
Initial speed (m/s)	7800	7800
Initial bearing (deg)	0	0
Initial flight path angle (deg)	-0.01	-2
Break-up altitude (km)	100	70

Table 8: SCARAB Correlation Initial Conditions

The simulations have been run from the initial conditions shown in Table 8 using both a spacecraft and an upper stage

with a mass of 1000kg and hypersonic continuum ballistic coefficient of 160kg/m². As this is an object-oriented correlation, a fixed fragmentation altitude has been used, and it has been found that the shallow and steep re-entries require substantially different break-up altitudes to obtain a reasonable match to the SCARAB results.

The set of components which have been tuned to the spacecraft re-entry results are given for each material in Table 9. It is worth noting that for reasonable results in comparison with SCARAB, a higher break-up altitude, and a higher ballistic coefficient range, than would be expected are required. This is partly due to the presence of a 'main fragment' of high ballistic coefficient which is often present in SCARAB runs due to the nature of the fragmentation model. A component for each combination of the ballistic coefficient and mass was run, and the total mass scaled to the total vehicle mass in order to calculate the average mass loss map and the landed mass fraction. A reasonably large component number was required to obtain a reasonable spread of the mass loss over the altitude range required. High ballistic coefficients were needed in order to push the heat pulse lower in the atmosphere, and thus to record mass losses below 50km, down to the 40km predicted by SCARAB.

	Aluminium	Copper	Steel	Titanium
Number of Components	48	40	32	25
Ballistic Coefficients (kg/m ²)	30, 50, 80, 100, 150, 200, 300, 400	50, 80, 100, 150, 250, 400, 500, 700	30, 50, 80, 100, 150, 200, 300, 400	30, 50, 80, 150, 300
Masses (kg)	10, 20, 50, 100, 150, 200	2, 5, 10, 20, 40	1, 2, 5, 10	0.2, 0.3, 0.5, 1, 2

Table 9: Components Used for Each Material

Although the components were selected to match the mass losses from the spacecraft simulations, it was decided to use the same components in the upper stage simulations, with the adjusted re-entry angle and break-up altitude.

The comparison of the mass loss in the spacecraft simulations with altitude for aluminium and copper is shown in Figure 1 and for steel and titanium in Figure 2. In general, the curve matches are good. The high altitude mass loss of aluminium in the SCARAB runs is not captured by the object oriented model, but the peaks in the mass loss are well represented for all the materials. In general, the matches are slightly conservative in that the mass loss is at lower altitudes in the SAM fits, but this is consistent with the overall approach.

As the original SCARAB simulations use a model for the major aluminium component (AA7075) which is known to be outdated [1], an improved aluminium material model has been used. This has the effect of shifting the mass loss to lower altitudes by a few kilometres and reducing the mass lost. The global emissions from SAM are within 20% of SCARAB in all cases.

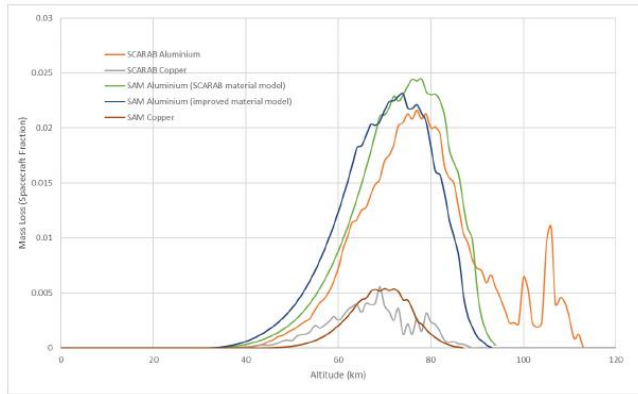


Figure 1: Spacecraft Mass Loss with Altitude (Aluminium and Copper)

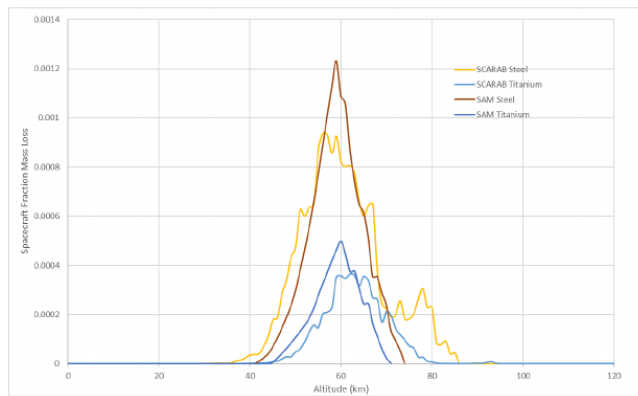


Figure 2: Spacecraft Mass Loss with Altitude (Steel and Titanium)

With the same components being run, and having been tuned to the spacecraft simulations, it should not be expected that the mass loss fit would be very good. Indeed, the aluminium mass losses are very high in comparison with the SCARAB results. It should be noted, however, that there is a very large uncertainty over the fragmentation processes of re-entering vehicles, and that the SCARAB melt-based model requires complete demise of a contiguous line of facets in order to trigger separation of fragments. This results in slower break up and produces a small number of large fragments, and thus a higher mass fraction of material reaches the ground in simulation. Models resulting in earlier catastrophic fragmentation tend to produce a larger number of component level fragments which leads to greater mass losses during re-entry. With this consideration, the limited SCARAB data set available and the fact that a higher mass loss to the environment is more conservative for this environmental application, these higher mass loss figures are deemed acceptable.

The construction of a representative spacecraft and upper stage has thus been performed using a set of equivalent mass and ballistic coefficient components for metallic materials.

However, many of the materials of interest are not metals. In order to capture the trajectories of these other material components, a set of fragments have been selected for each of the materials of interest. The resulting trajectories are used as an input for the chemistry assessment for these materials, with the exception of glass where the mass loss by shear will be calculated by the existent SAM model. The components considered for each material are given in Table 10. Note that relatively high ballistic coefficients are used to account for the potential for components to demise whilst still attached to a larger fragment. Glass is assumed to shear and form particles, and so explicit masses can be used. For the other materials, a thickness of material is assumed.

	Ballistic Coefficient (kg/m ²)	Mass (kg)	Thickness (mm)
Glass	30, 50, 80, 150, 300	0.2, 0.5	N/A
Cyanate Ester Resin	30, 50, 80, 150, 300	N/A	1.5
Carbon (incl. Fibres)			
PET / PTFE / PFA / Polypropylene	30, 50, 80, 150, 300	N/A	1.5
GFRP	30, 50, 80, 150, 300	N/A	1.5
Cork-Phenolic	30, 50, 80, 150, 300	N/A	1.5

Table 10: Components Used for Each Material

For the metallic species and glass, for which a direct mass loss calculation can be made with reasonable reliability, a map of the particulate mass loss can be constructed. The components of Tables 9 and 10 (for glass) are run over a number of trajectories. The trajectories are generated parametrically with the variation given in Table 11. The variability in the re-entry is driven by the ballistic coefficient and the re-entry angle of the parent spacecraft at 120km. The remaining values are as in the SCARAB comparison analysis given in Table 8. The same values are used for both the spacecraft and the upper stage analyses.

Parameter	Uncontrolled Re-Entry	Controlled Re-entry
Parent Ballistic Coefficients (kg/m ²)	80, 120, 160, 200, 250	80, 120, 160, 200, 250
Parent Re-entry Angle (deg)	0, -0.005, -0.01, -0.015, -0.02	-1, -1.25, -1.5, -1.75, -2
Break-up Altitude (km)	100	70

Table 11: Trajectory Parameter Variation

It is worth noting that this requires 25 trajectory simulations per component ballistic coefficient for both the controlled and uncontrolled re-entries. With five ballistic coefficients being considered for most of the species, this gives 250 trajectories for each species. Indeed, for the metallic and glass objects, the melt is performed explicitly during the simulation, so each mass is also treated separately resulting in a minimum of 500 (and up to 2400) trajectory computations for each of these species.

The mass loss during a controlled spacecraft re-entry has been calculated in a 2D map of along track distance and altitude, similarly to the previous analysis. As the initial condition at 120km is controlled in this case, the along track distance is measured from this location. This provides mass loss maps for all the particulate species of interest.

5. PARTICULATE EMISSIONS

The assessment of the chemistry has been made, for the most part, using the assumption that the species can be treated independently, and that there is no significant interaction between the products.

As the spacecraft is heated, there are a number of key chemical processes that vary substantially with the different species present. Metallic materials tend to undergo surface oxidation, but this oxide layer is generally quite thin. Once the metal melts, it is encased in an oxide 'bag' [1]. This bag is then expected to shear, resulting in smaller droplets whose size is driven by the local shear environment. These droplets will again form a surface oxide layer. This is the major source of particulates. Production of particles by shear is also expected from glass-like materials, where the material failure into droplets is driven by the reduction of viscosity with temperature. As glassy materials do not have a melt temperature a different physical process is considered. There is also some possibility of particulates being formed by the destruction of weak carbonaceous chars which are the products of the chemistry of organic components.

The process of relatively large liquid droplets undergoing a secondary breakup, which is the most likely phenomenology of the particulate producing emissions from the above discussion, is well studied in the field of sprays for droplets. This analysis is generally restricted to conditions where the density ratio between the liquid and gas phases, and the Reynolds number based on the droplet diameter are sufficiently high. These requirements are easily fulfilled in this application. The key parameters involved in the fragmentation process are the aerodynamic forces which tend to cause the breakup, and the surface tension forces which act to hold the droplet together. The controlling parameter is the Weber number

$$We = \frac{\rho_g V^2 d_l}{\sigma}$$

where V denotes the relative velocity between the liquid and gas phases.

Assessing the likely behaviour in re-entry is not simple as the conditions are supersonic, but the Weber numbers are small due to the low ambient density. Therefore, there is no directly applicable correlation. A number of approaches have been assessed and the correlation of Faeth [2], which relates the Sauter Mean Diameter (SMD) of the resultant distribution to the critical size of droplet which is unstable, selected as the most conservative in that the smallest droplet sizes are produced.

The minimum Weber number of breakup is $We=12$ [3], which provides the limiting droplet size at a given trajectory condition. From Faeth, the resultant SMD is 9% of this value. Using a typical metallic droplet density of 5000kg/m^3 and a surface tension of 1N/m , the size range of the droplets

produced is from $50\mu\text{m}$ to over $500\mu\text{m}$. Separate altitude based size maps have been produced for each particulate species type.

It is worth noting that these droplet sizes are relatively high and would not be expected to have a long residence time in the atmosphere. Therefore, it is expected that metallic particulate production has limited impact on the environment.

Particulates produced from carbonaceous chars are likely to be a larger issue. The majority of the understanding of the spallation from carbonaceous chars in high speed airflows comes from the field of Thermal Protection Systems (TPS), where the heatshield has to be designed to withstand the heat fluxes and pressures such that spallation does not occur.

The spallation from the charred organic residue can take two forms. Either it can break from the char and be ejected as a relatively large lump, of a few hundred microns to a few millimetres in size, or it can be similar to the particles which form smoke, which are significantly smaller in size. As the small particles are much more relevant to the environmental impact due to their larger surface area per unit mass and their longer residence time, it is this second production process which is of particular interest.

As the data on the mass loss and particle size in the re-entry context is limited, the literature on the dynamics of fires was investigated. Usefully, Mulholland [4] provides a detailed description of the particulate mass production and the sizes of the particles produced by a range of different materials. Interestingly, the smoke conversion factor, which is the initial mass fraction that is converted to particulate smoke has a maximum value of 0.19, suggesting that no more than approximately one-fifth of the initial char mass should be considered as producing small particles. These data have been produced for free burning at surface atmospheric conditions, whereas the ventilation can strongly affect the smoke production. In flight conditions, the availability of oxygen to the surface is not inhibited by the environment due to the rapid motion of the vehicle. Therefore, there is expected to be sufficient oxygen available for relatively free combustion of the material to occur. It is worth noting that in transport, it is expected that the amount of smoke will be reduced due to deposition on other surfaces.

The peak carbon particle mass fractions, and the Sauter Mean diameters given by Mulholland for different species of interest are given in Table 12. The conversion factor is the mass of particles produced per unit mass of 'fuel'. Separate values are given for pyrolysis and oxidation conditions for some materials. In general, the values are about 10% of the initial mass, if it can be considered that the entire mass reacts. The mean diameters are of the order of 1 micron, with those produced in surface ablation (oxidation) being smaller than those produced in the pyrolyzing phase. In order to err slightly on the side of conservatism, a particle size SMD of 0.5 microns is selected for use in the atmosphere simulations.

Some basic analysis of inorganic materials, such as alumina and silicate paints, was performed which suggested that no emissions of interest were produced by these materials.

6. GASEOUS EMISSIONS

The precise chemical composition of ablation products is usually less important to the re-entry engineer than the overall energy balance, which determines the effectiveness of the TPS. This means that detailed chemistry is generally less well reported than the overall rate of de-densification and energy balance. Therefore, the analysis of the organic chemistry is necessarily simplified.

The trajectories calculated for the components of Table 10 are used as input for this analysis. Once again, 250 trajectories are assessed per input species considered. As the demise of these objects is calculated post-trajectory, there is no coupling between the trajectory and the demise, and therefore the ballistic coefficient is not changed during the trajectory calculation. As this results in the ballistic coefficient being maintained for longer than might be expected, and thus the trajectory penetrating deeper into the atmosphere at higher speeds, this can be considered a conservative assessment from an emissions point-of-view.

A first estimate of the chemistry in the shock layer can be produced using standard equilibrium chemistry codes, such as the NASA Chemical Equilibrium with Applications (CEA) code. Closed system analysis principally provides information concerning the likely split of species produced by ablation. However, this split can be a strong function of temperature. Temperature and pressure are input parameters to closed system analysis, so need to be estimated.

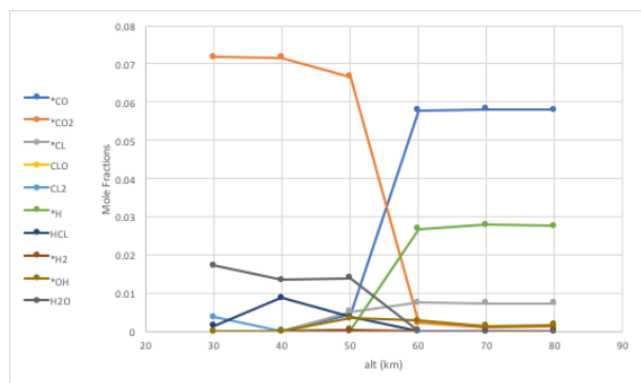


Figure 3: Species moles fraction from CFRP equilibrium ablation

Equilibrium chemistry principles can also be used to calculate pressures and temperatures behind a normal shock to place sensible bounds on the input parameters to be used for equilibrium chemistry calculations. Chemical equilibrium

calculations are generally more strongly a function of temperature than pressure. The temperature to be used for equilibrium chemistry calculations is difficult to specify as it will depend on how far from the surface ablation products are transported by diffusion. For the purposes of this analysis, a temperature which is half way between the equilibrium post shock temperature and a typical cold wall value of 300K is used. The ablation products obtained from a cyanate ester resin CFRP are shown in Figure 3.

Note that the majority product species are carbon dioxide and water at low temperature, but that at high temperature hydrogen and chlorine radicals are present. The chlorine radicals are of particular interest for the environmental impact, although it should be noted that the emission can occur at relatively high altitudes.

In order to provide a mass loss map for the CFRP material, the FGE ablation code FABL is run along the trajectories in Table 11. The basic phenomenology observed is that pyrolysis occurs at relatively high altitudes. Surface ablation of the remaining char then takes place at lower altitudes. As discussed in Ramirez et al. [5] chlorine bearing species are usually released along with the pyrolysis gases. As such, chlorine bearing species should be produced at relatively high altitudes. At some point during the entry, the CFRP becomes fully charred and no more pyrolysis gas is produced. The char does not begin to ablate until the surface temperature has increased to a level sufficient to promote oxidation. This happens later in the trajectory as can be seen in the example shown in Figure 4.

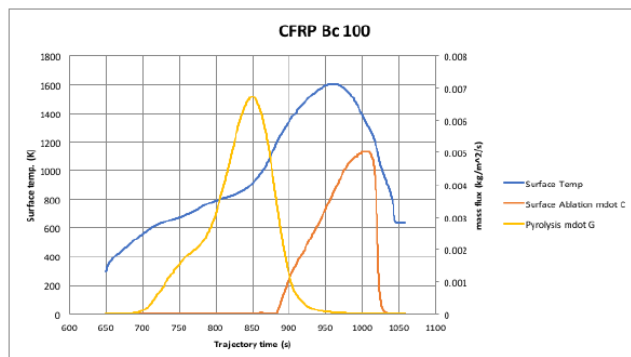


Figure 4: Surface Temperatures and Ablation Rates for CFRP

Similarly to the analysis for the metallic particles, a 2D map of the mass loss is found as a function of the along track distance and altitude, and scaled to the fraction of the initial spacecraft mass. The fraction of the input species mass loss is calculated from the remaining mass fraction of the 1.5mm thickness layer of ablative material across the simulations.

The mass losses from the CFRP material are separated out into the pyrolysis mass loss, which is representative of the

cyanate ester resin, and the char mass loss, which is representative of the carbon material.

To produce the emissions from the organic species, the mass loss maps must be separated into the relevant species mass fractions. Investigation has shown that the equilibrium mass fractions from both the pyrolysis and char mass losses have been very consistent through the trajectory and across trajectories. Therefore, the equilibrium mass fractions at the peak mass loss have been selected as representative in order to distribute the mass loss into emissions, both inclusive and exclusive of oxygen being available, as shown in Table 12.

Species	Pyrolysis		Char	
	No Oxygen	Oxygen Present	No Oxygen	Oxygen Present
CO	0.180		0.156	
CN			0.144	
H ₂	0.028		0.010	
C ₂ H	0.003		0.030	
C ₃	0.031		0.658	
HCl	0.669	0.615		
CO ₂		0.376		3.129
H ₂ O		0.231		0.120
Cl ₂		0.040		

Table 12: CFRP Emissions

An improvement on the emissions produced can be obtained from the study of the finite rate chemistry in the shock layer and the wake of a debris object. The organic chemistry analysis provides the equilibrium species concentrations at the surface of the object undergoing demise – these are the values where oxygen is not available from Table 12. Two phenomena can possibly be captured by a CFD calculation; the impact of the air chemistry in the shock layer, which is expected to be the major source of NO_x, and the relaxation of the chemistry. This is essentially the oxidation of the ablation products as the temperature falls in the wake, which could provide the best indication of the OH emissions. There is also the possibility that the emissions will interact, and therefore, these simulations can also account for this. These effects will provide a delta on the species emissions produced in the above analysis to give the final values to be used in the atmosphere simulations.

Four cases have been run, as shown in Table 13. All computations have been performed with MISTRAL, a three-dimensional Navier-Stokes code with several flow models.

Case	Ballistic Coefficient (kg/m ²)	Trajectory Point	Altitude (km)
1	100	Peak char ablation	45.8
2	100	Peak pyrolysis emission	82.5
3	50	Peak char ablation	55.4
4	50	Peak pyrolysis emission	81.4

Table 13: CFD Cases

A large domain grid which runs for approximately 500 radii behind the debris object was used in order to capture the chemical relaxation from the shock layer. Two stages of grid refinement have also been used to ensure that grid convergence was achieved.

At the higher altitudes, where the pyrolysis dominates, the species of interest diffuse to about 1.5 radii from the centerline in the wake, but at the lower altitudes, the species of interest are concentrated within 0.5 radii of the centerline.

There are two key aspects of interest. Firstly, the chlorine radical mass fraction is significant downstream, such that it is not all converted to HCl as in the equilibrium assessment. Further, there is a large amount of NO produced in the shock layer due to the dissociation of air molecules in the shock wave. This demonstrates that a key aspect of the impact of the re-entry may not, in fact, come from the materials of the spacecraft itself.

The complete emission per unit mass of the incident spacecraft or upper stage is given in Table 14.

	Uncontrolled				Controlled			
	Spacecraft		Upper Stage		Spacecraft		Upper Stage	
Total Emissions	Best Guess	Worst Case	Best Guess	Worst Case	Best Guess	Worst Case	Best Guess	Worst Case
Aluminium Particles Mass	0.315	0.346	0.533	0.586	0.237	0.346	0.401	0.586
Titanium Particles Mass	0.008	0.040	0	0	0.028	0.052	0	0
Copper Particles Mass	0.070	0.075	0	0	0.059	0.075	0	0
Steel Particles Mass	0.020	0.060	0.045	0.136	0.013	0.060	0.028	0.135
Glass Particles Mass	0.007	0.062	0.001	0.007	0.032	0.062	0.003	0.007
Carbon Particles Mass	0.004	0.041	0.004	0.029	0.002	0.041	0.003	0.029
Carbon Dioxide Mass	0.091	0.022	0.064	0.015	0.062	0.022	0.044	0.015
Carbon Monoxide Mass	0.018	0.372	0.013	0.259	0.012	0.372	0.00	0.259
Water Mass	0.003	0.011	0.004	0.016	0.003	0.011	0.003	0.016
HCl Mass	0.007	0.008	0.005	0.005	0.007	0.008	0.005	0.005
OH Mass	0.002	0.005	0.001	0.003	0.002	0.005	0.001	0.003
Cl Mass	0.015	0.015	0.010	0.011	0.014	0.015	0.010	0.011
NO Mass	0.350	0.400	0.350	0.400	0.350	0.400	0.350	0.400
O Mass	0.350	0.400	0.350	0.400	0.350	0.400	0.350	0.400
NH ₃ Mass	0.0005	0.003	0	0	0.0005	0.003	0	0

Figure 14: Summary Emissions Mass Production

7. CONCLUSIONS

The mapping of the mass loss from re-entering spacecraft to a relevant set of emissions for use in atmospheric models is a complex process. This has been achieved via the use of a set of representative trajectories on vehicles where the material composition is properly defined. Particles produced are unlikely to have a major effect on the atmosphere as they are relatively large, and gaseous emissions of interest are likely to be limited to chlorine species from cyanate ester resins and nitrogen oxides from shock layer chemistry.

8. REFERENCES

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